

Electric Vehicle Adoption and Charging Infrastructure in Maryland

DATA 205 Final Report

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Project question	Is charging infrastructure keeping up with EV growth in Maryland?
Final model	$EV_per_Capita \sim Stations_per_1000 + Median_Income$
Unit of analysis	23 Maryland counties
Main takeaway	Income was the stronger predictor after adjusting for population; infrastructure gaps remain in high-demand counties.

Introduction and Project Overview

Electric vehicles are becoming more common across Maryland, but the growth is not happening evenly across every county. Some counties have large EV totals and strong demand, while others are still much lower. This project looks explores the question: is charging infrastructure keeping up with EV adoption, and what factors help explain the differences?

At first, I planned to compare total EV registrations with total charging stations. As I worked through the data, I realized that raw totals could be wrong because Maryland counties have very different population sizes. A larger county will naturally have more people, more vehicles, and more chargers. Because of that, my final analysis uses `EV_per_Capita` as the outcome and `Stations_per_1000` as a standardized measure of charging access. I also included `Median_Income` because EV ownership is connected to affordability, access to home charging, and local resources.

The goal of this project is not to prove that charging stations or income directly cause EV adoption. Instead, the goal is to find patterns that can help explain where EV adoption is strongest, where infrastructure gaps may be, and how Maryland counties differ from each other.

Datasets Used

This project uses four county-level datasets. The first dataset is Maryland EV registration data from MDOT/MVA. This dataset was used to measure EV adoption by county and over time. The second dataset is the Alternative Fuel Station dataset from the U.S. Department of Energy, which includes charging station locations and charger information. The third dataset is the Maryland Median Income from the NIH which includes median household income by county, and the fourth dataset Maryland Population by County from data Montgomery includes Maryland county population.

Together, these datasets allowed me to compare EV adoption, charging access, income, and population-adjusted rates. The final cleaned dataset has one row per county and includes variables such as `EV_per_Capita`, `Stations_per_1000`, `Median_Income`, total EV registrations, charging stations, and EV-to-charger ratio.

Goals, Tools, and Cleaning

The main goal was to build a clear data science story about EV adoption in Maryland. More specifically, I wanted to clean and merge multiple datasets, compare counties fairly, create useful visualizations, and build a multilinear regression model that connects EV adoption with charging access and income.

I completed the analysis in Python using Google Colab. Pandas was used for cleaning, merging, and creating new variables. Matplotlib was used for the visualizations. Statsmodels was used for the multilinear regression model and regression assumption checks. GitHub was used to organize the notebooks, images, cleaned data, and report materials.

The cleaning process included converting street addresses to their corresponding county names, filtering to EV records, grouping charging station data by county, merging datasets by county, and creating new variables. The most important created variables were `EV_per_Capita`, `Stations_per_1000`, and EV-to-charger ratio. These helped make the analysis fairer because counties have very different population sizes and infrastructure needs.

Exploratory Data Analysis and Data Story

The EDA graphs helped shape the story before building the regression model. The first major pattern is that EV adoption has increased over 88% in 2 years in Maryland, especially in large counties. This growth is what makes the infrastructure question important.

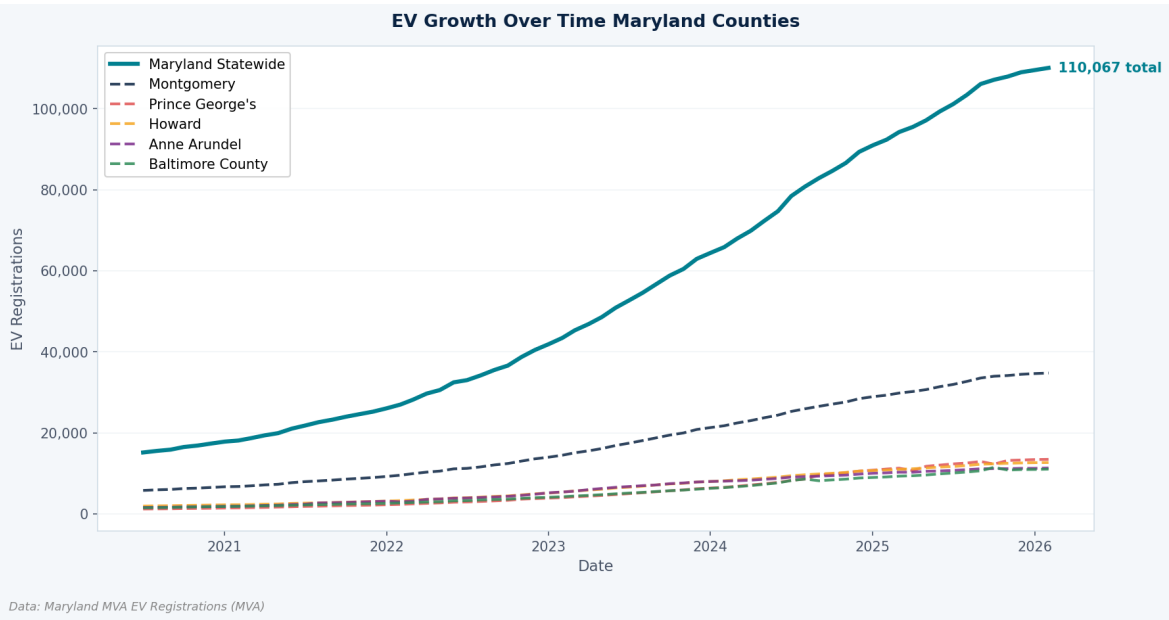


Figure 1. EV growth over time in top 5 Maryland counties. The upward trend shows why charging infrastructure has become a major planning question.

The second major pattern is that charger pressure is not evenly distributed. Counties with more EVs can still experience a shortage if charging stations do not grow at the same pace. The EV-to-charger ratio helps show where demand may be higher relative to available infrastructure. Something interesting about this graph is for being third largest county, Baltimore has one of the lowest EV-to-charger ratios in Maryland. This is because Baltimore County is a very dense area allowing the government to install lots of chargers in apartment complexes, parking lots, and public areas. Another reason is because the government is getting ready for when EV adoption starts to grow rapidly in Baltimore.

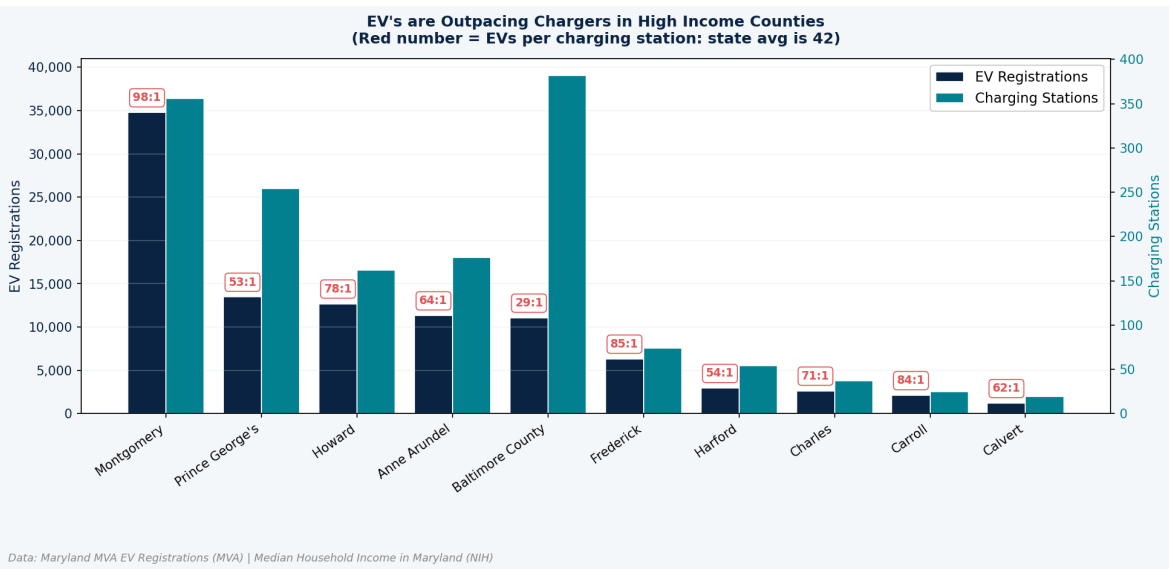


Figure 2. EVs and charging stations in high-income counties. Counties with stronger EV adoption can still show infrastructure pressure when EV totals grow faster than station counts.

The third EDA graph connects directly to the final model. It shows that higher income counties often have higher EV demand and may also face larger charger shortages. This supported the decision to include Median_Income

in the regression model instead of only looking at charging stations. The red line is the state average of 42.5 EV per one station. High income counties have almost double that at 76.5 EV per station.

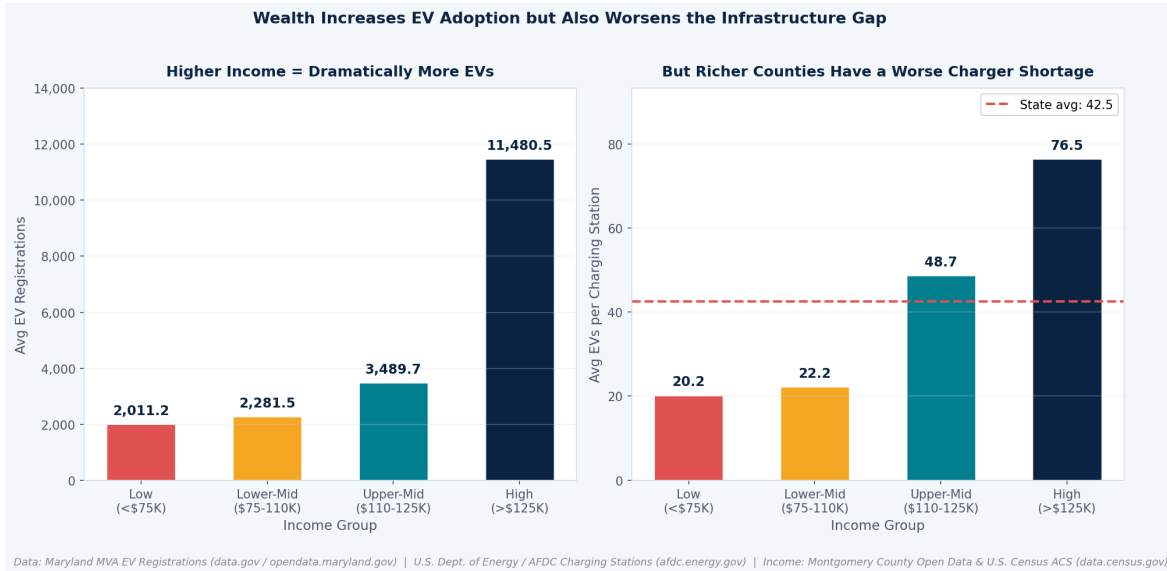


Figure 3. Income and charging shortage by county. This graph supports the idea that income is connected to EV adoption and infrastructure demand.

Final Data Product and Regression Model

The final dataset is a cleaned county-level dataset, a set of EDA visualizations, and a multilinear regression model. The cleaned dataset brings the main variables together in one place, which makes it easier to compare counties and explain the project story clearly.

My final multilinear regression model was $EV_per_Capita \sim Stations_per_1000 + Median_Income$. I used EV_per_Capita so that counties could be compared more fairly. I used $Stations_per_1000$ to measure charging access after adjusting for population. $Median_Income$ was included because EV adoption can be influenced by the ability to afford an EV and access reliable charging.

```
=====
MODEL: EV_per_Capita ~ Stations_per_1000 + Median_Income
=====
R²                : 0.685
Adjusted R²       : 0.654
β (Stations_per_1000) : +0.011450
β (Median Income)   : +2.72e-07
p (Stations_per_1000) : 0.1478
p (Median Income)   : 0.0000
RMSE              : 0.005200 EVs per capita
=====
```

Figure 4. Final regression model summary from the analysis notebook.

The model explained about 68.5% of the variation in EV adoption rate across counties. $Stations_per_1000$ had a positive relationship with EV_per_Capita , but it was not statistically significant in this model because of the high p-value. $Median_Income$ had a positive and statistically significant relationship, which means income was the stronger predictor after adjusting for population. This does not prove that income causes EV adoption because 31.5% is still caused by other factors, but it does show a strong county-level pattern.

Regression Assumption Checks

Linearity: Test used: residuals vs fitted values plot.

Result: the residuals were mostly centered around zero with no major curved pattern.

Meaning: the linear model is reasonable for showing the general relationship between EV_per_Capita, Stations_per_1000, and Median_Income. Some counties still have larger prediction errors, so the model does not explain every county perfectly.

Normality of residuals: Test used: Q-Q plot of standardized residuals.

Result: most points followed the diagonal line, with few points deviating at the upper end.

Meaning: the residuals are mostly normal, so the model results are usable. The deviations show that a few counties behave differently from the main pattern.

Equal variance: Test used: residuals vs fitted plot and Breusch-Pagan test.

Result: the residual plot did not show a major funnel shape, and the Breusch-Pagan test was included as the formal check for unequal spread.

Meaning: the errors are reasonably consistent across fitted values, so there is not a strong equal variance problem in the model.

Independence: Test used: county-level study design and Durbin-Watson check.

Result: each county was treated as one separate observation, and the Durbin-Watson check was used as a basic residual independence check.

Meaning: this assumption is reasonable for county-level data, but nearby counties may still share regional patterns.

No strong multicollinearity: Test used: Variance Inflation Factor.

Result: VIF was used to check whether Stations_per_1000 and Median_Income were too highly related.

Meaning: the predictors did not show a major overlap concern, so each variable can be interpreted as adding separate information to the model.

The residuals vs fitted values plot below supports the linearity and equal variance checks. Most residuals are centered around zero, but there are still some larger errors, which means some counties are not predicted as well as others.

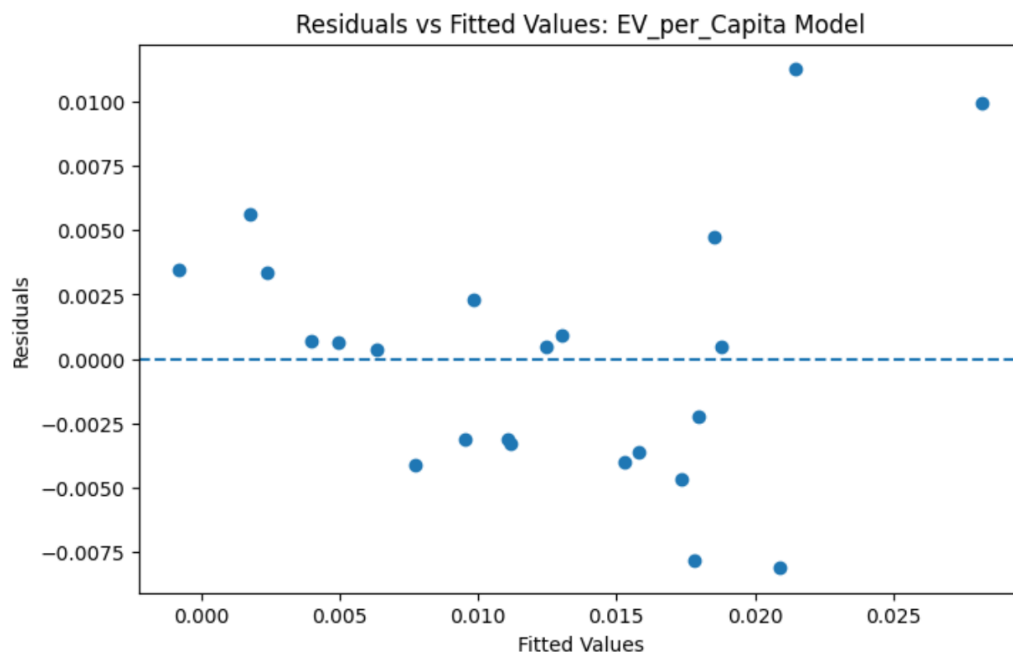


Figure 5. Residuals vs fitted values for the EV_per_Capita regression model.

The Q-Q plot of standardized residuals below supports the normality check. Most points follow the diagonal line, while a few points deviate at the upper end. Overall, the assumptions were reasonably met, but the small sample size of 23 counties means the results should be interpreted as patterns rather than causal proof.

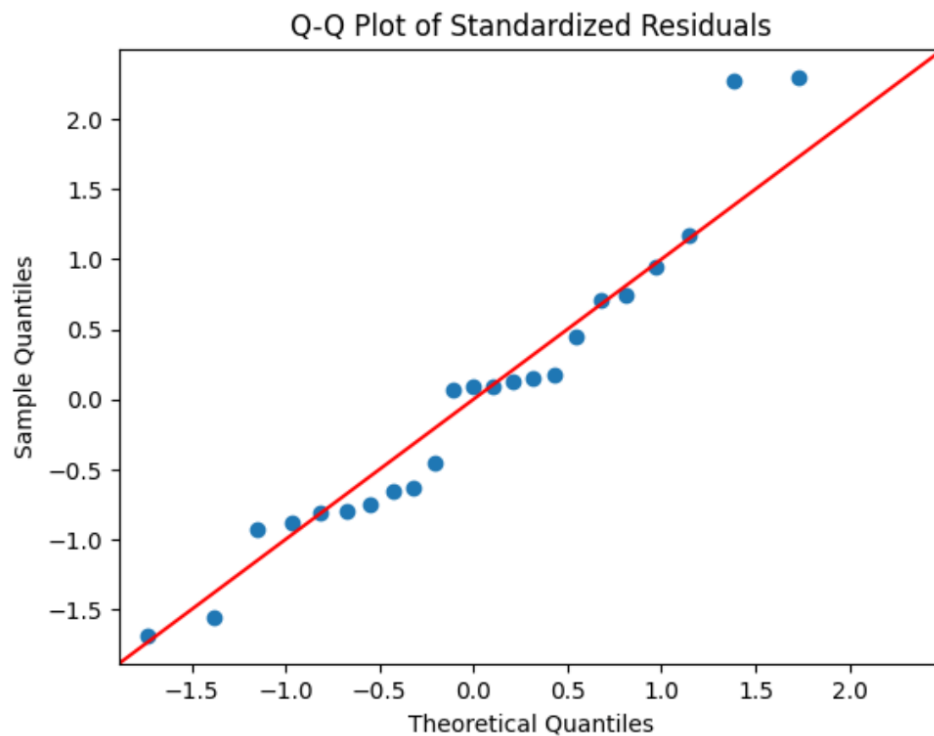


Figure 6. Q-Q plot of standardized residuals used to check residual normality.

For equal variance, I also used the Breusch-Pagan test. For independence, each county was treated as one separate observation, while recognizing that nearby counties may still share regional patterns. For multicollinearity, I used VIF to check whether Stations_per_1000 and Median_Income overlapped too much. These checks helped make the model more reliable and transparent.

Challenges and Reflection

One of the biggest challenges was merging the datasets cleanly. County names and formats were not always consistent, so I had to clean and standardize the data before the merge worked correctly. Another challenge was deciding how to compare counties fairly. Raw EV totals were useful for showing scale, but they did not fully account for population differences.

The small sample size was also a major limitation. Since Maryland has only 23 counties, the regression model has limited statistical power. This means that the model can support the story, but it's a perfect prediction tool. A few counties, especially large counties like Montgomery County, can have a strong effect on the model.

The most rewarding part was turning the raw data into a clear story. The project started with separate files, but by the end I had a cleaned dataset, visuals, a regression model, and a practical interpretation of what the results mean for Maryland EV infrastructure.

Next Steps and Recommendations

Next time, I would like to make a predictive model of the data. My plan is to forecast EV adoption through 2030 to help guide infrastructure planning. I would also add more variables such as population density, home ownership, commute distance, and urban versus rural classification. These could help explain EV adoption more completely. From a planning perspective, counties with fast EV growth should pay attention to infrastructure pressure, not just total charger counts. More Level 2 charging in residential, workplace, and public areas could be especially useful in high-demand counties where EV adoption is growing quickly.

Acknowledgements

I would like to thank Professor Alraee for her guidance and feedback throughout this project. Her comments helped me improve the model, include regression assumption checks, and explain the limitations more clearly. I

would also like to thank Professor Saidi, Professor Peri, my Data Science cohort, and Montgomery County Government/Data Montgomery for support and data resources.

References

Maryland Department of Transportation Motor Vehicle Administration.

U.S. Department of Energy.

U.S. Census Bureau and Data Montgomery.

Maryland Department of Environment.